

## 1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

### Definition 1.10.1

The **Kronecker product** of an  $m \times n$  matrix  $\mathbf{A}$  and a  $p \times q$  matrix  $\mathbf{B}$  is the  $mp \times nq$  block matrix  $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$ . The **vec operator** places the  $n$  columns of an  $m \times n$  matrix one below another, giving a vector of length  $mn$ :  $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$ .

### Proposition 1.10.1

For matrices of compatible sizes:

- (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$  (mixed-product rule);
- (b)  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$ , and  $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$  for nonsingular  $\mathbf{A}, \mathbf{B}$ . More generally  $\mathbf{A}^- \otimes \mathbf{B}^-$  is a generalized inverse of  $\mathbf{A} \otimes \mathbf{B}$ ;
- (c)  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A})\text{tr}(\mathbf{B})$  and  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A})\text{rank}(\mathbf{B})$ ;
- (d) if  $\mathbf{A}$  ( $m \times m$ ) and  $\mathbf{B}$  ( $p \times p$ ) are symmetric with eigenvalues  $\lambda_i$  and  $\mu_j$ , the eigenvalues of  $\mathbf{A} \otimes \mathbf{B}$  are the products  $\lambda_i\mu_j$ , and  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p(\det \mathbf{B})^m$ ;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f)  $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$  and  $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$ .

*Proof.*

- (a) The  $(i, j)$  block of the left side is  $\sum_k a_{ik}c_{kj}\mathbf{B}$ , which is the  $(i, j)$  block of the right side. (b) Transposition is direct, and the rest follows from (a), for example  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ .
- (b) With  $\mathbf{A} = \mathbf{PAP}^\top$  and  $\mathbf{B} = \mathbf{QMQ}^\top$ , (a) and (b) give  $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{A} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$ , a spectral decomposition because  $\mathbf{P} \otimes \mathbf{Q}$  is orthogonal and  $\mathbf{A} \otimes \mathbf{M}$  is diagonal with entries  $\lambda_i\mu_j$ . The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$  is idempotent with the same rank as  $\mathbf{A} \otimes \mathbf{B}$  (Proposition 1.3.3(a)). Its trace is  $\text{tr}(\mathbf{AA}^-)\text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A})\text{rank}(\mathbf{B})$  by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The  $j$ th column of  $\mathbf{ABC}$  is  $\mathbf{ABc}_j = \sum_k c_{kj}\mathbf{Ab}_k$ , which is the  $j$ th block of  $(\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$ . The trace identity is  $\sum_{ij} a_{ij}b_{ij}$  written two ways.

■

The determinant formula in (d) holds for all square  $\mathbf{A}$  and  $\mathbf{B}$ , not only symmetric ones. We need only the symmetric case.

**Example. (The balanced two-way layout)**

Let observations  $y_{ij}$ ,  $i = 1, \dots, a$ ,  $j = 1, \dots, b$ , be stacked with  $j$  running inside  $i$ . The additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$  has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because  $\mathbf{I}_a \otimes \mathbf{1}_b$  has a 1 in column  $i$  exactly for the  $b$  observations in row  $i$ . With  $\bar{\mathbf{J}}_k = k^{-1}\mathbf{J}_k$ , a symmetric idempotent averaging matrix, the mixed-product rule shows that  $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ ,  $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$  and  $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  replace each observation by its row mean, its column mean and the grand mean. The matrix  $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  is symmetric and idempotent, with trace  $a + b - 1 = \text{rank}(\mathbf{X})$ . It is the projection onto  $\mathcal{C}(\mathbf{X})$  (Theorem 16.1.2 in Chapter 16). The script checks this for  $a = 3$  and  $b = 4$ , together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")          # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))          # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),          # intercept
               np.kron(np.eye(a), one(b)),      # row indicators
               np.kron(one(a), np.eye(b))])     # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

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